



Original software publication

NeuroSpeech

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ARTICLE INFO

Article history:

Received 4 August 2017

Accepted 4 August 2017

Keywords:

Speech processing

Dysarthria

Parkinson's disease

Phonation

Articulation

Prosody

Intelligibility

Python

Software

ABSTRACT

NeuroSpeech is a software for modeling pathological speech signals considering different speech dimensions: phonation, articulation, prosody, and intelligibility. Although it was developed to model dysarthric speech signals from Parkinson's patients, its structure allows other computer scientists or developers to include other pathologies and/or measures. Different tasks can be performed: (1) modeling of the signals considering the aforementioned speech dimensions, (2) automatic discrimination of Parkinson's vs. non-Parkinson's, and (3) prediction of the neurological state according to the Unified Parkinson's Disease Rating Scale (UPDRS) score. The prediction of the dysarthria level according to the Frenchay Dysarthria Assessment scale is also provided.

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Code metadata

Current code version	v1.1
Permanent link to code/repository used for this code version	https://github.com/ElsevierSoftwareX/SOFTX-D-17-00058
Code Ocean compute capsule	https://codeocean.com/2017/07/30/neurospeech-colon-an-open-source-software-for-parkinson-apos-s-speech-analysis/code
Legal Code License	MIT
Code versioning system used	none
Software code languages, tools, and services used	Python, C++
Compilation requirements, operating environments & dependencies	Anaconda 3, python 3.6, Praat 6029, ffmpeg
If available Link to developer documentation/manual	https://github.com/ElsevierSoftwareX/SOFTX-D-17-00058
Support email for questions	jcamilo.vasquez@udea.edu.co

Software metadata

Current software version	v1.1
Permanent link to executables of this version	For example: https://github.com/jcvasquezc/NeuroSpeech/tree/master/src/release/PDTool.exe
Legal Software License	MIT
Computing platforms/Operating Systems	Windows
Installation requirements & dependencies	Anaconda 3, Python 3.6, Praat 6029, ffmpeg
If available, link to user manual – if formally published include a reference to the publication in the reference list	https://github.com/jcvasquezc/NeuroSpeech/tree/master/doc
Support email for questions	jcamilo.vasquez@udea.edu.co

DOI of original article: <http://dx.doi.org/10.1016/j.dsp.2017.07.004>.

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<http://dx.doi.org/10.1016/j.softx.2017.08.004>

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1. Motivation and significance

This software was designed to model pathological speech signals considering different speech dimensions: phonation, articulation, prosody, and intelligibility, and speech tasks: sustained vowels, isolated words, sentences, read texts, and monologues. *NeuroSpeech* is based on state-of-the-art signal processing techniques which have been successfully used for modeling dysarthric speech signals. The current version of the software is based on recordings of Parkinson's patients and healthy speakers; however, its structure allows other scientists to include other pathologies and methods; thus this software is a significant contribution to the topic of pathological speech processing. Additionally, the structure of the software allows clinicians, care givers, and patients, to use it in order to follow their speech therapy. To the best of our knowledge, this is the first software with the characteristics described above, and we consider that it will help other researchers to contribute to the state-of-the-art in pathological speech assessment from different perspectives, e.g., from the clinical point of view for interpretation, and from the computer science point of view enabling the test of different measures and pattern recognition techniques.

2. Impact

NeuroSpeech allows the analysis of speech signals considering four different dimensions: phonation, articulation, prosody, and intelligibility. This differentiated analysis is new and will allow the study of different speech pathologies that impact different aspects of the speech production process. Each of those dimensions are modeled in this software considering state-of-the-art methods, which have been tested and validated in previous experiments and publications. These characteristics of the software make it very original and flexible, allowing to address different pathologies. For instance, in dysarthria it is relevant to study phonation, articulation, and prosody, separately; in laryngeal cancer, the study of phonation is very important; and in hypernasality it is relevant to study articulation.

Besides the possibility to address different speech pathologies, *NeuroSpeech* allows the developer or computer scientist to train the system with labels of different scales. The current version is trained with the UPDRS (Unified Parkinson's Disease Rating Scale) and the m-FDA (modified – Frenchay Dysarthria Assessment) scales, but

it can be re-trained with other scales depending on the specific interest of the researcher.

NeuroSpeech is currently being used by Grupo de Neurociencias de Antioquia (GNA) at University of Antioquia in Medellín¹ during the clinical consultations to Parkinson's patients (or other patients suffering from diseases that produce dysarthria). Additionally, in order to promote the use of the software, it was uploaded to <https://github.com/jcvasquezc/NeuroSpeech> and will be also uploaded to the ISCA (International Speech Communication Association) website.

3. Conclusions

A new software for the analysis of pathological speech signals is presented. The software is based on state-of-the-art methods that have been validated in previous experiments and publications. We have presented a configuration of the software to analyze voice recordings of Parkinson's patients; however, the same platform can be easily adapted to perform the analysis of other voice pathologies like dysphonia or hypernasality. Four different aspects or dimensions of speech can be modeled using *NeuroSpeech*: phonation, articulation, prosody, and intelligibility. The analysis of pathological speech signals based on such a “segmentation” by dimensions allows the user to make specific conclusions regarding different aspects of the speech production process. This characteristic makes *NeuroSpeech* a very good option for clinicians, patients, and care givers to assess pathological voices.

This software is freely accessible and open source, and to the best of our knowledge, this is the first attempt to launch an easy to use software, freely accessible and open source to help people of different nature (clinicians, patients, or scientists) to perform their own analyses and to make their own conclusions regarding different aspects of speech pathologies.

Acknowledgments

This project was funded by COLCIENCIAS (grant # 111556933858), and also by CODI (grant # 2015-7683) at Universidad de Antioquia. Additionally, this work was partially developed during the 2016 Jelinek Memorial Summer Workshop on Speech and Language Technologies, which was supported by Johns Hopkins University via DARPA LORELEI, Microsoft, Amazon, Google, and Facebook.

¹ <http://neurociencias.udea.edu.co/>.